

Department of Computer Science and Engineering

Individual Contribution Report — Team Assignment

To be completed by EACH student individually and submitted along with the team's assignment. This report explains what YOU specifically did, so it must be written in your own words and reflect only your own work.

A. Assignment & Team Information

Course Code & Name	CSA0614 – Design and Analysis of Algorithms
Assignment Title	Conflict-Free Exam Timetable Scheduling Using Backtracking (Graph Colouring)
Team / Group No.	3
Team Members (all names)	Haresh Kumar N L (192425009), Nirmal Kasi (192424248), Mohan Kumar J (92424060)
Date of Submission	02/09/2026

B. Student's Own Details

Student Name	Haresh Kumar N L
Register / Roll No.	192425009
Role in the Team	Algorithm Design and Implementation Lead.

C. Task-wise Individual Contribution

List every task/module you personally worked on. Add rows if needed. Do not describe work done by teammates.

S.No.	Task / Module Handled	Description of Work Done	Tools / Techniques Used	Approx. Hours
1	Problem Understanding and Graph Formulation	Converted the exam scheduling problem into a graph-colouring model by representing courses as vertices and conflicts as edges.	Graph Modelling, Graph Colouring	4
2	Plain Backtracking Algorithm	Developed the basic recursive backtracking approach for assigning colours while checking conflicts.	Backtracking, Recursion	5
3	Pruning-Enhanced Backtracking	Worked on selecting constrained vertices to reduce unnecessary search during colouring.	Pruning, Most-Constrained Vertex	5
4	Pseudocode and Algorithm Design	Prepared pseudocode and structured the steps for both colouring approaches.	Algorithm Design, Pseudocode	3
5	Core Algorithm Implementation	Implemented the graph-colouring logic, colour assignment, conflict checking and backtracking in JavaScript.	JavaScript, Recursion	6

D. Team Contribution Percentage Summary

To be filled identically by every team member so contributions are consistent and sum to 100%. Disagreements should be discussed and resolved before submission.

Team Member Name	Register No.	% Contribution	Signature
Team Member Name	Register No.	% Contribution	Signature
Haresh Kumar N L	192425009	35%	Haresh Kumar
Mohan Kumar J	192424060	35%	Mohan Kumar
Nirmal Kasi	192424248	30%	Nirmal
Total			100%

E. Course Outcomes / Skills Applied

Course Outcome(s) Addressed by MY Work	How I Applied It
CO2 – Algorithm Design and Problem Solving	I applied algorithmic problem-solving by converting the exam timetable scheduling problem into a graph-colouring problem. Each course was represented as a vertex and a conflict between two courses was represented as an edge. This representation allowed the scheduling constraints to be handled systematically using graph algorithms.
CO3 – Application of Algorithmic Techniques	I applied Backtracking to construct a valid colouring. The algorithm assigns a colour to a course, checks whether the assignment violates any existing conflict, and recursively proceeds to the next course. If no valid colour is available, the algorithm reverses the previous assignment and tries another possibility.

F. Challenges Faced and How I Resolved Them

Describe any technical or teamwork challenges you personally faced, and the steps you took to resolve them.

One of the major challenges I faced was understanding how the real-world exam scheduling problem could be represented using graph theory. Initially, the problem could be viewed simply as assigning time slots to courses, but this does not directly show the relationship between courses that share students. I resolved this by treating every course as a vertex and every conflict as an edge. This made it possible to apply Graph Colouring directly.

Another challenge was implementing recursive Backtracking correctly. Since the algorithm repeatedly assigns colours and then returns to earlier decisions when a conflict occurs, it was important to maintain the correct state of the assignments. I resolved this by ensuring that a colour was assigned only after checking its validity against already-coloured neighbouring vertices. When an assignment failed, the colour was removed before returning to the previous recursive level.

Implementing the pruning enhancement also required careful consideration. The purpose of pruning is to reduce unnecessary search, but the optimization should not remove valid solutions. I therefore maintained the same safety-checking condition used by the basic algorithm while changing the order in which vertices were considered.

Testing was another challenge because the algorithm needed to work for both $k=3$ and $k=4$. I checked that valid assignments were produced when enough colours were available and that the algorithm correctly identified cases where the available number of colours was insufficient.

G. Individual Learning Outcome / Reflection

What did YOU personally learn from completing your part of this assignment? How did it build on classroom concepts?

This assignment significantly improved my understanding of Backtracking and Graph Colouring because I had to move from theoretical algorithm steps to an actual working implementation. I learned that Backtracking is essentially a systematic search process in which decisions are made incrementally and invalid decisions are removed when they lead to a dead end.

I also gained a better understanding of recursive algorithms. Before implementing the solution, recursion could be viewed mainly as a theoretical programming concept. During this project, I understood how recursive calls can represent different levels of a search tree and how returning from a recursive call effectively performs the backtracking operation.

The pruning component helped me understand the importance of optimization in algorithm design. A correct algorithm is not necessarily an efficient algorithm. Reducing the number of unnecessary assignments can make a significant practical difference, especially as the number of courses and possible colours increases.

Overall, my main learning outcome was the ability to connect graph theory, recursion, constraint checking and algorithm optimization to a practical scheduling problem.

H. Declaration

I declare that the contribution described above accurately represents the work I personally performed for this team assignment, and that I have not misrepresented the work of any other team member as my own.

Student Signature	Date
Haresh Kumar N L	09/02/2026